Chapter 3
Vectors and cloning

I. Introduction

II. Vectors

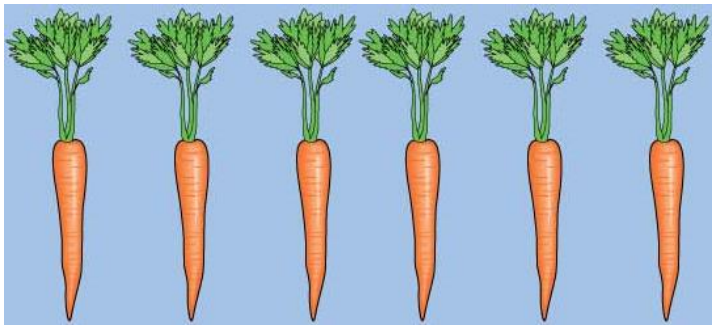
III. Enzymes in cloning

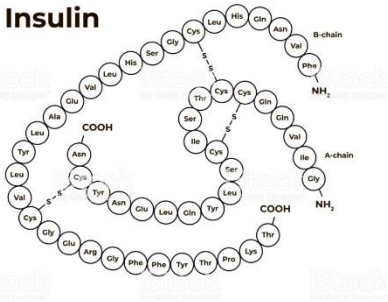
IV. Cloning process

I. Introduction

Clone – Cloning

- **Clone (dòng):** a clone in biotechnology is any biological material (DNA, molecule, cell, or whole organism) that is genetically identical to the source from which it originated.



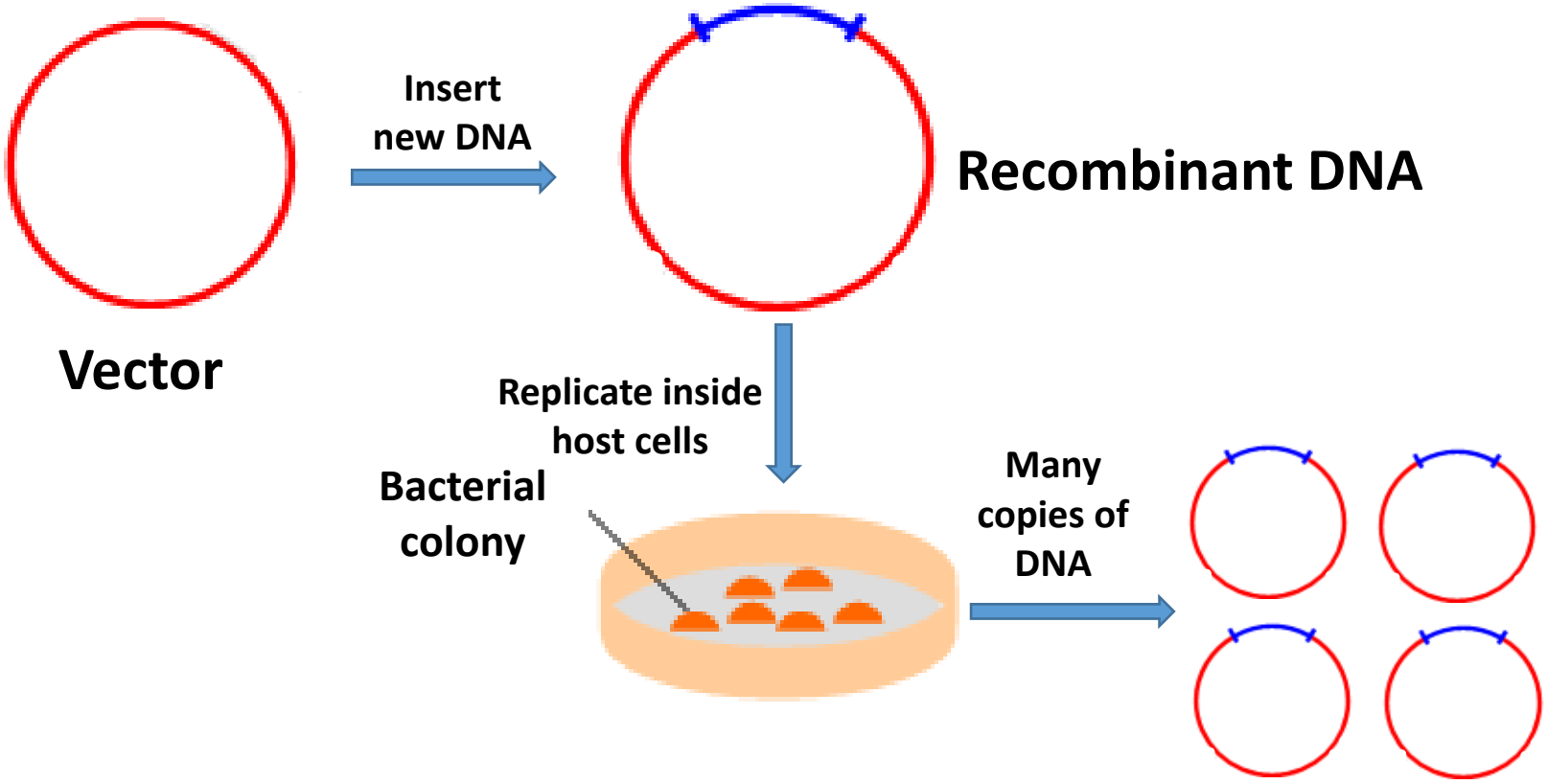






Clone – Cloning

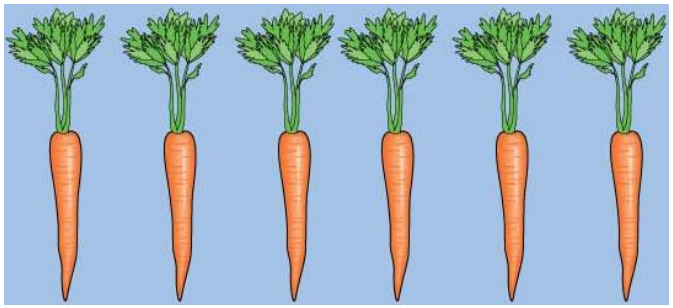
- **Cloning (KT tạo dòng):** the cloning process is defined as the technique used to create genetically identical copies of a DNA fragment, cell, or organism.
- The process typically includes the following steps: Isolation of DNA of interest → Insertion into a vector → Introduction into a host cell → Replication and selection → Screening and expression.

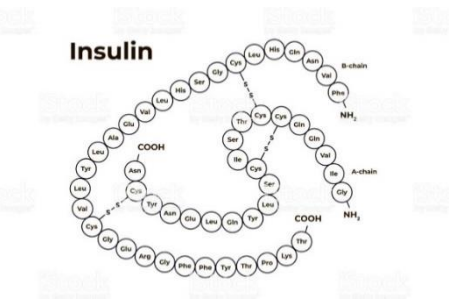


Clone – Cloning

The term can apply at different levels:

- **Molecular cloning**: identical copies of a specific DNA fragment or molecule, propagating it in a host organism such as bacteria.
- **Cellular cloning**: producing a population of identical cells derived from a single parent cell.
- **Organismal cloning**: creating an entire multicellular organism that is genetically identical to another.

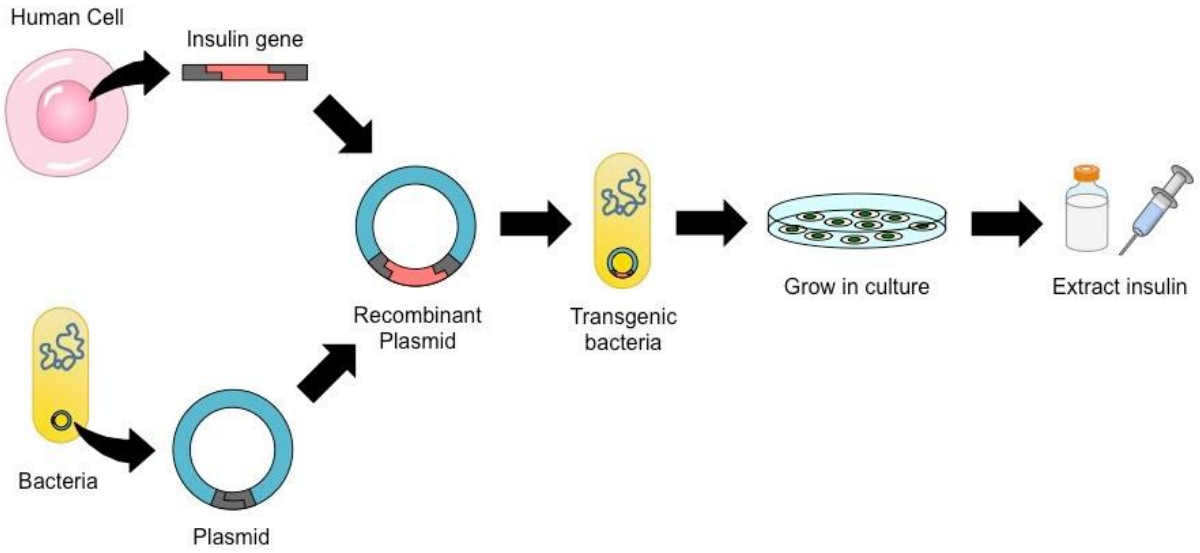






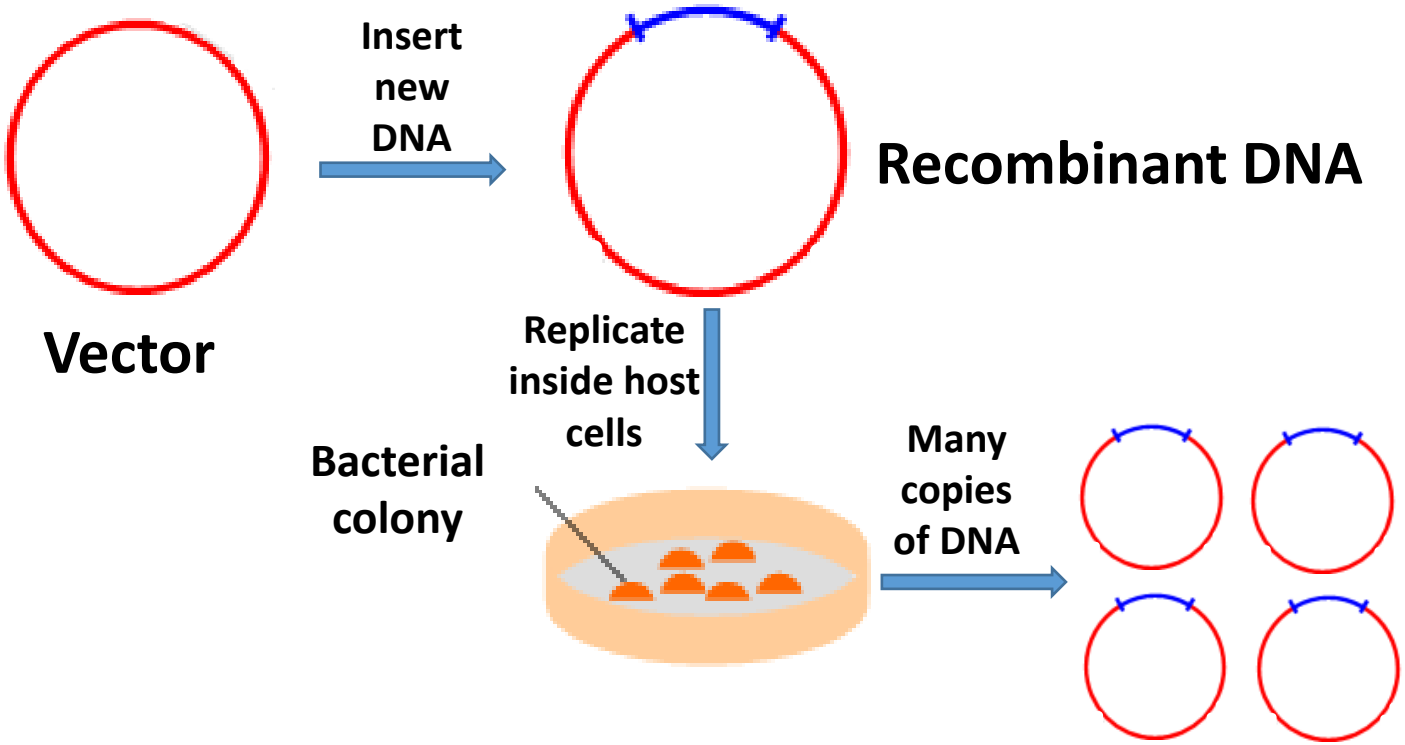
Meanings of cloning

- Producing gene or molecular clones with:
  - desired characteristics
  - large amount
  - in control
- Applied in research, medicine, agriculture, and industry



Basic components of a cloning process

- Vector
- Enzymes (RE, ligase)
- Recombinant DNA (by PCR,…)
- Host cell



II. Vectors

Cloning vector?

A cloning vector is a small piece of DNA used as a carrier to transfer a specific DNA fragment into a host cell for replication.

Key features of a cloning vector:

- **Origin of replication (ori):** allows the vector to replicate independently inside the host cell.
- **Selectable marker genes:** typically antibiotic resistance genes (e.g., ampicillin resistance) that help identify host cells containing the vector.
- **Multiple cloning site (MCS):** a short DNA sequence with several unique restriction enzyme sites for inserting foreign DNA.
- **Small size:** makes it easy to manipulate and transfer into host cells.

II. Vectors

Common types of cloning vectors:

- **Functions:**
  - Cloning vector
  - Expression vector
- **Source of vector:**
  - Plasmids (most widely used)
  - Phage vectors
  - Artificial vectors: YACs, BACs,…

II.1. Plasmid vector

‘Natural’ plasmid in bacterial cell

- A small, circular, double-stranded DNA molecule that exists independently of the bacterial chromosomal DNA.
- Replication: Contains its own origin of replication (ori), allowing independent replication from the bacterial chromosome.
- Copy number: Can exist in multiple copies inside a single bacterial cell (1 -100 copies/cell)
- Functions: antibiotic resistance,…

Plasmid vectors

Generation 1st

- pSC101 by Herbert Boyer và Stanley Cohen in 1973
- Size: ~9 kb
- Copy number: Low-copy number (~1-5 copies per cell).
- Selectable marker: tetracycline resistance (Tet^R) gene.
- *Disadvantages*

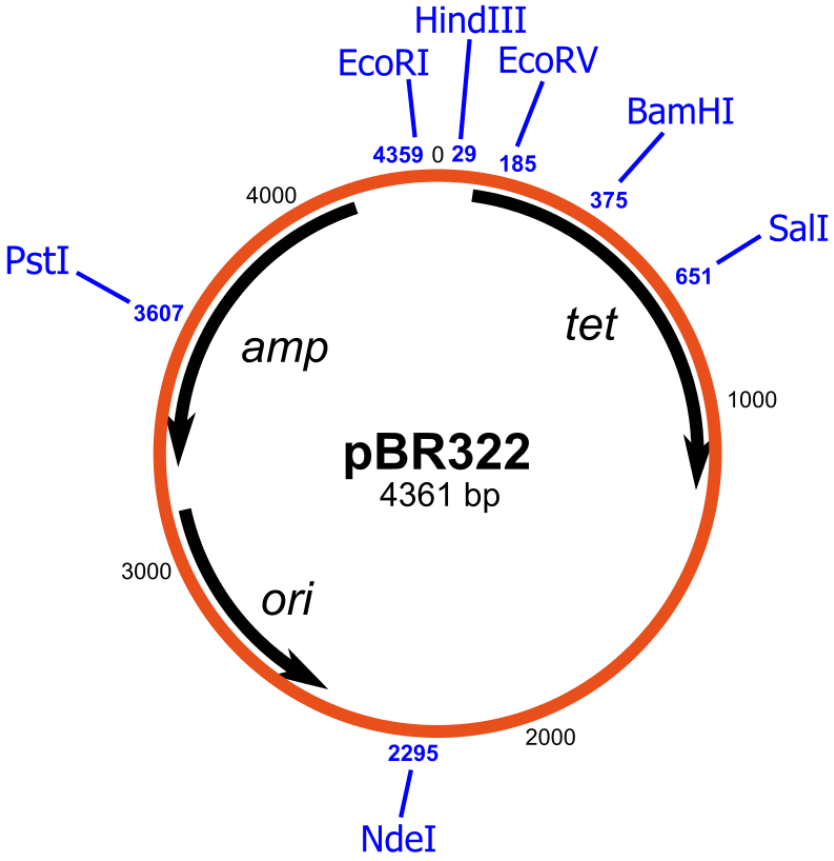
  - large size (compared to next plasmid vector generations)
  - low-copy number
  - one RE recognition site

Plasmid vectors

Generation 2nd

- pBR322 by Bolivar and Rodriguez in 1977.
- Length 4,361bp; 15-20 copy/cell
- Multiple RE recognition sites
- Selectable marker: tetracycline, ampicillin resistance genes
- ***Disadvantages***

  screening steps complicated



Commercial plasmid vectors

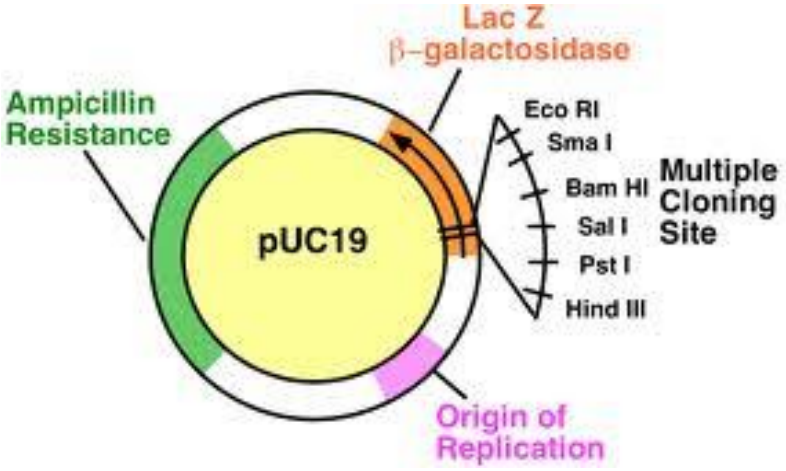
Generation 3rd

- **Cloning plasmid vector**

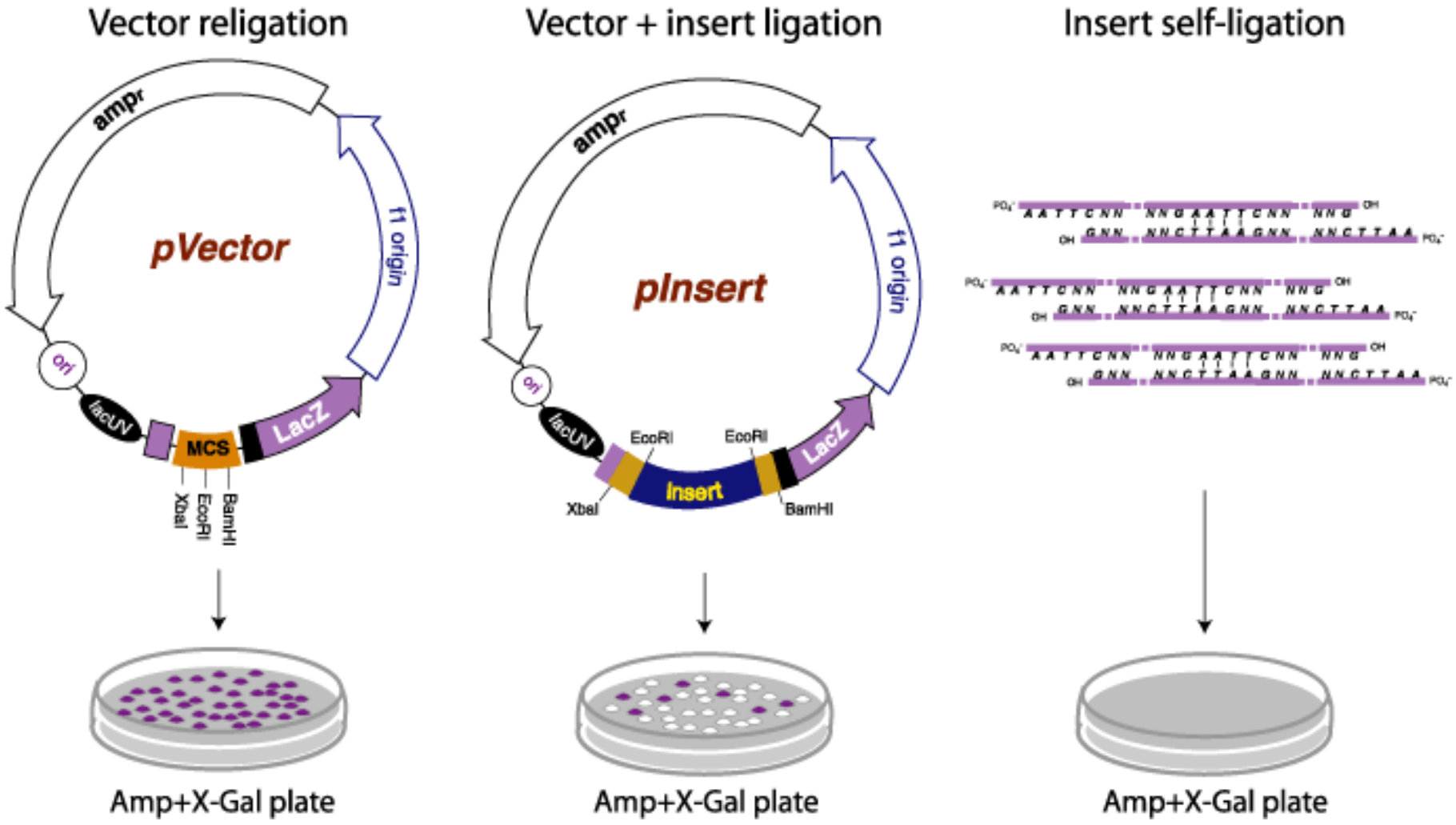
  + pUC (pUC18, pUC19)

  + lacZα gene (part of the lac operon): allows colony-color based screening…

  + Multiple Cloning Site (MCS) – synthetic sequence with many unique restriction enzyme sites for easy DNA insertion.



Screening based on *lacZ* gene



Commercial plasmid vectors

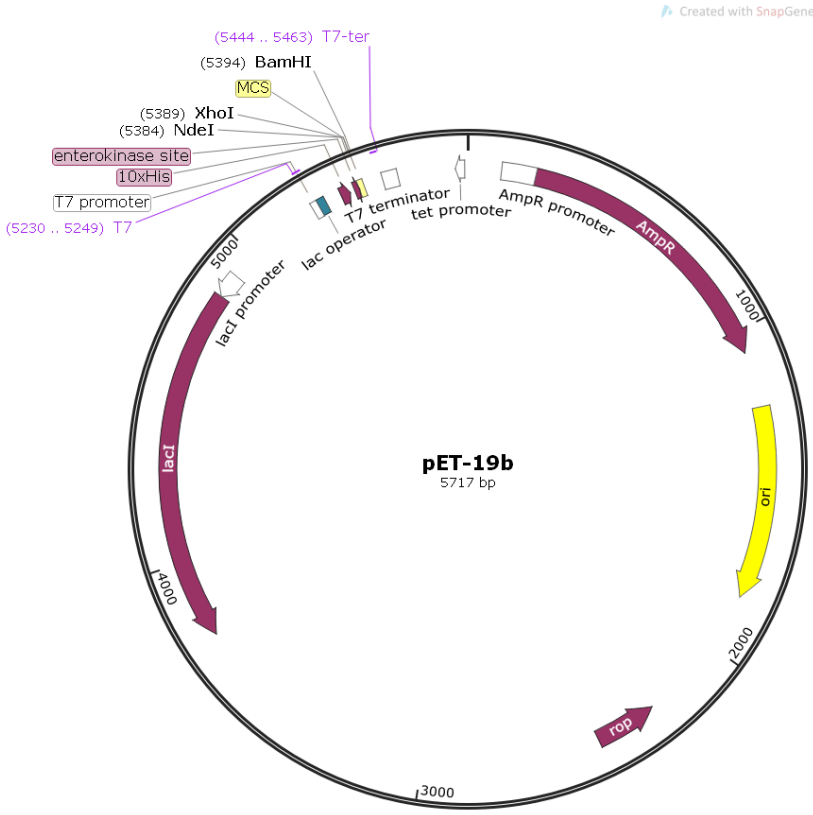
Generation 3rd

- **Expression plasmid vector**

  + pET, pGEM,…

  + Promoter supports transcription

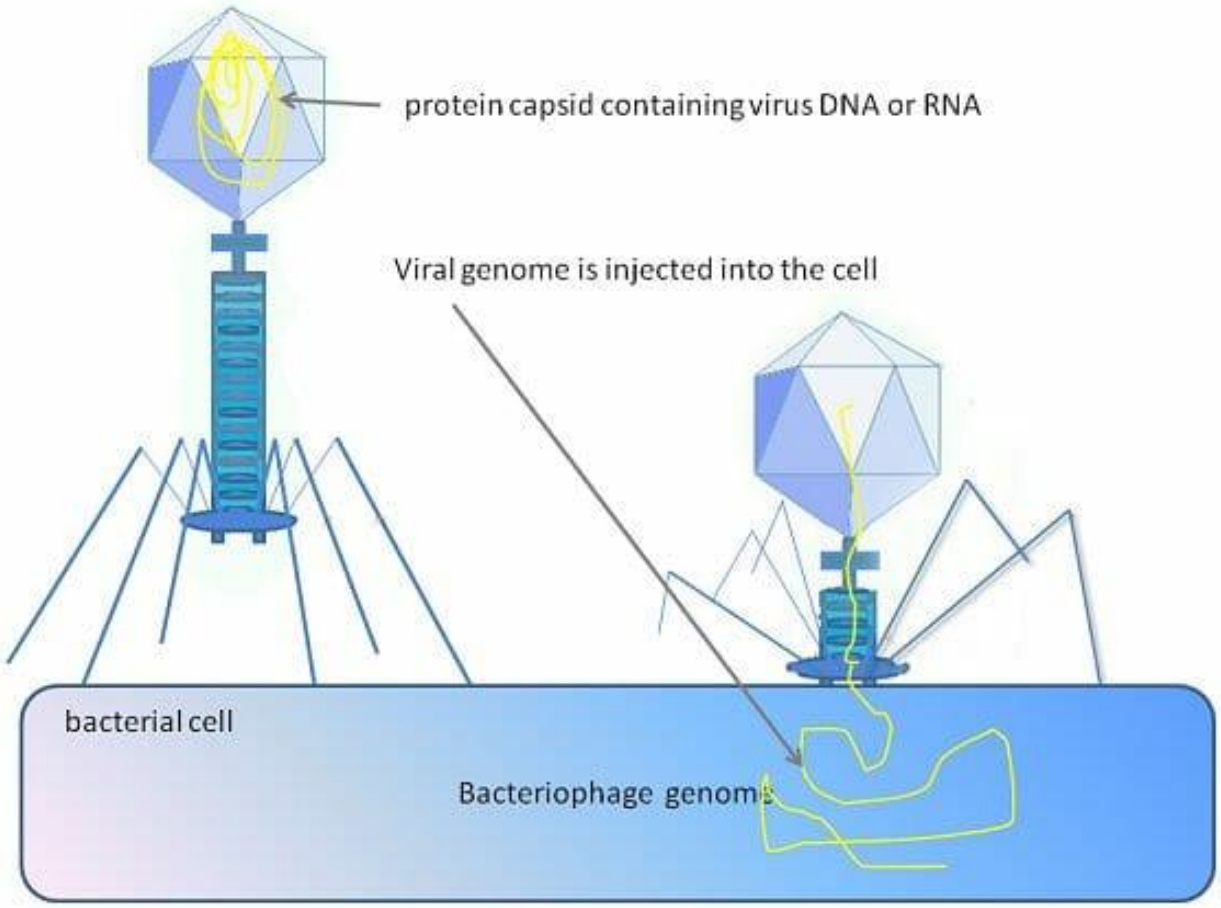
of recombinant DNA



II.2. Phage vector

- **What is phage?**

- virus infect bacteria

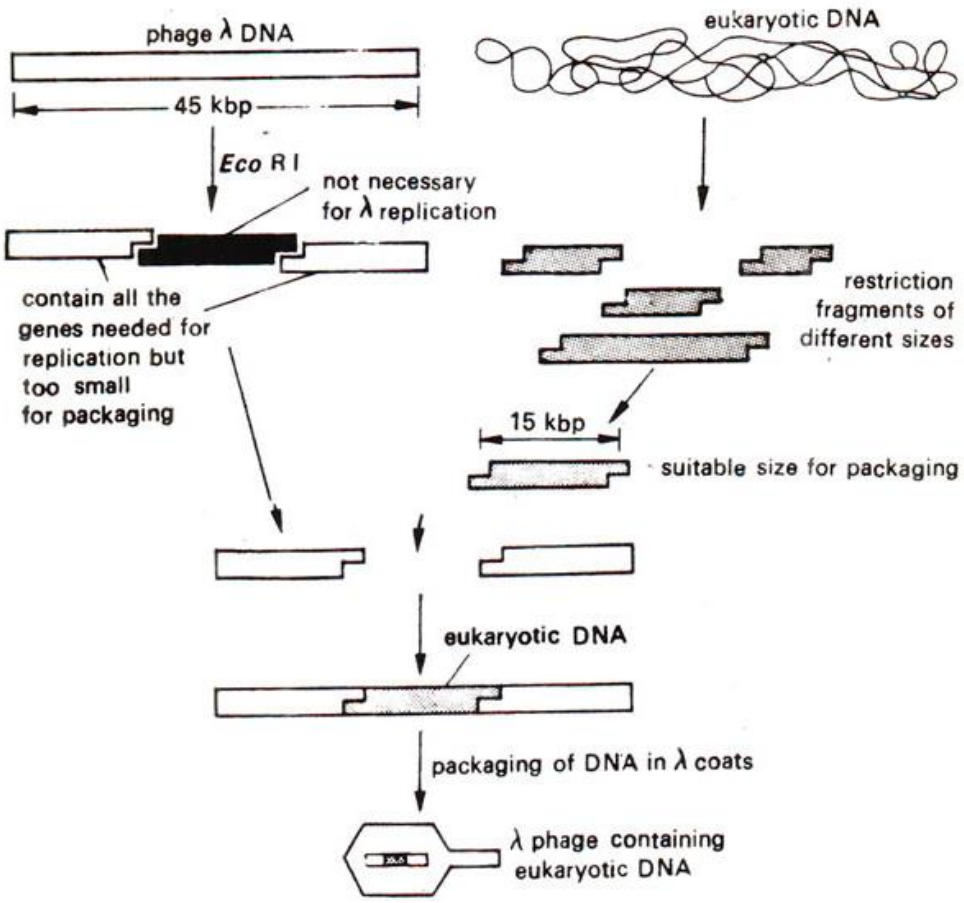


Phage lambda vector

The λ phage vector works by inserting foreign DNA into a modified λ genome → packaging into phage heads → infection of *E. coli* → replication → screening plaques to recover recombinant clones.

Advantages

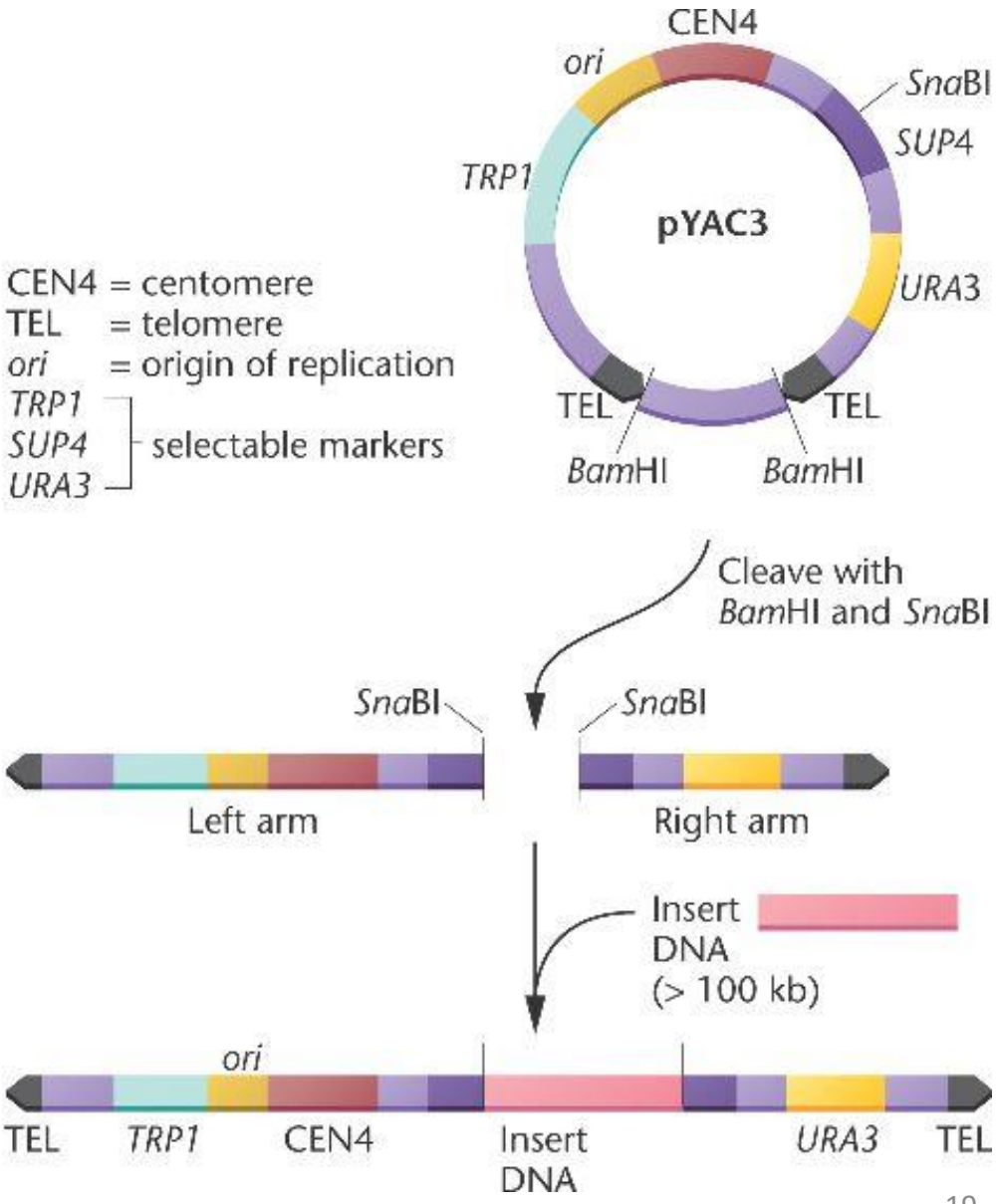
- Rapid replication in host cells
- Carry foreign fragment up to 23 kb



II.3. Yeast Artificial Chromosomes (Nhiễm sắc thể nhân tạo của nấm men YAC)

Essential elements for replication and division of yeast chromosomes : 2 **TEL** (telomere), 1 **CEN**, (centomere), 1 **ARS** (autonomously replicating sequence)

- Vector YAC includes 2 **TEL**, 1 **CEN**, 1 **ARS** (ori) and selectable markers
- A foreign fragment (up to 1000 kb) is inserted to 2 arms



III. Enzymes in cloning

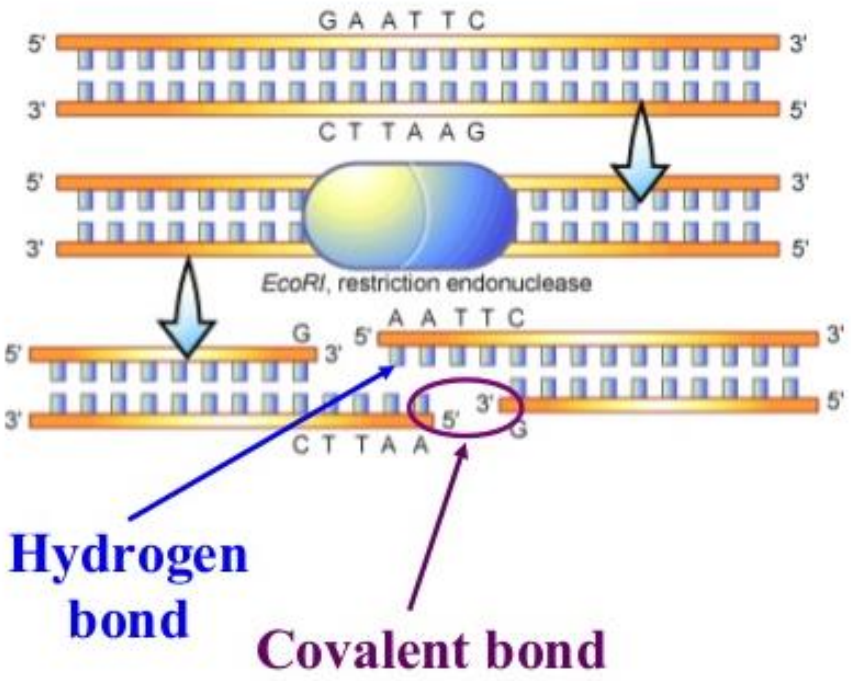
III.1. Restriction enzyme

III.2. Ligation enzyme (DNA ligase)

III. Enzymes in cloning

III.1. Restriction enzyme

A restriction enzyme (also called a restriction endonuclease) is a protein that cuts DNA at specific recognition sequences.

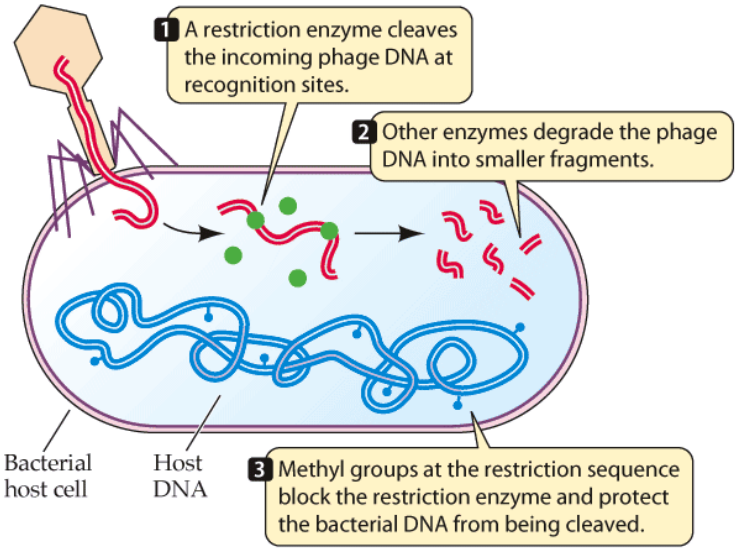


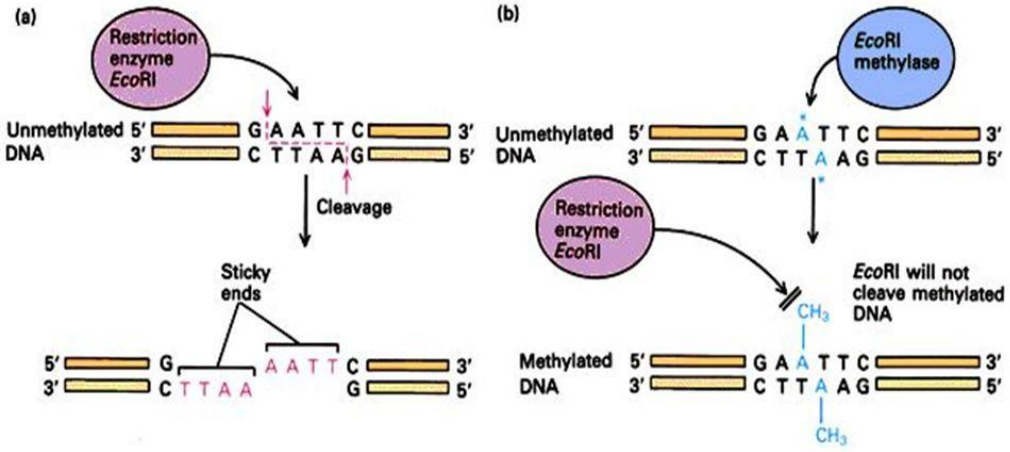
“restriction”??!!

III. Enzymes in cloning

III.1. Restriction enzyme

- **Roles in living cells**
  - “Immune system” of bacterial cells

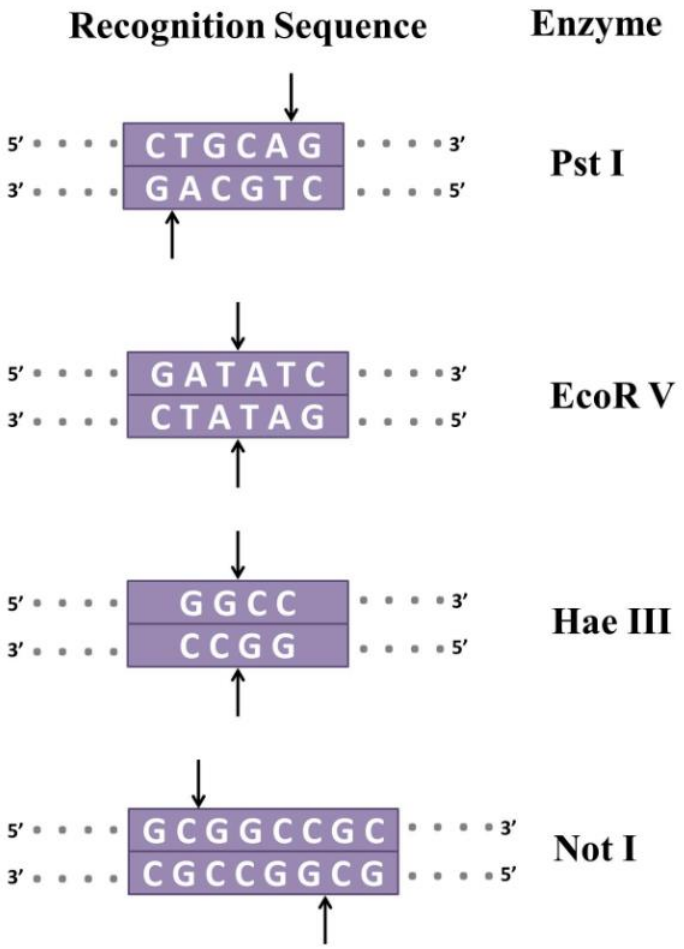




(Source: Purves's *Life: The Science of Biology, 7th Edition*)

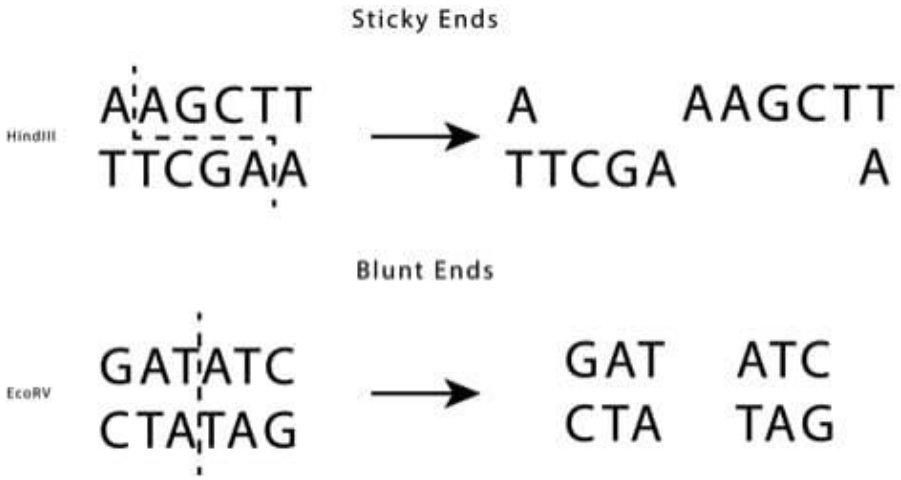
Recognition sequence

- Each restriction enzyme recognizes a specific short DNA sequence (often 4–8 base pairs long)
- Most recognition sequences are palindromic, meaning the sequence reads the same in the 5′ → 3′ direction on both DNA strands.
- In cloning, REs with 6-nu recognition sequence are often used…



Recognition sites

- Each restriction enzyme cuts at or near recognition site.
- Blunt end (đầu bằng), sticky end (đầu dính)



Name of RE

- Based on its sources

Name	Source Microorganism
Bam HI	*Bacillus amyloliquefaciens*
Eco RI	*Eschericia coli* RY13
Hind III	*Haemophilus influenzae* Rd
Not I	*Nocardia otitidis-caviarum*
Pst I	*Providencia stuartii*
Sma I	*Serratia marcescens*

Classification of restriction enzymes

Type I

- Complex enzymes
- Cut DNA at random sites far from the recognition sequence
- Rarely used in genetic engineering because cleavage is unpredictable.

Classification of restriction enzymes

Type II

- The most common and useful in molecular biology.
- Cut DNA at specific sites within or close to the recognition sequence.
- Examples: *Eco*RI, *Bam*HI, *Hind*III.
- Generate sticky ends or blunt ends, making them essential for cloning, recombinant DNA, and mapping.

Classification of restriction enzymes

Type III

- Cut DNA a short distance (about 20–30 bp) away from the recognition site.
- Recognition site is usually non-palindromic.

Classification of restriction enzymes

Type	Cutting Site	Usage
I	Random, far from recognition site	Rarely used
II	At/near recognition site	Most used in biotechnology
III	20–30 bp away from recognition site	Limited use

RE activity

- **Unit of RE**

1 U - the amount of enzyme required to completely digest 1 microgram (µg) of substrate DNA (usually λ DNA or plasmid DNA) in 1 hour under optimal conditions specified by the manufacturer (e.g., buffer, temperature, reaction volume)

Unit Definition of EcoRI - NEB
One unit is defined as the amount of enzyme required to digest 1 µg of λ DNA in 1 hour at 37°C in a total reaction volume of 50 µl.

1X NEBuffer™ EcoRI
100 mM Tris-HCl
50 mM NaCl
10 mM $MgCl_2$
(pH 7.5)

- **Inactivation of RE**
  - Why?
  - How?

Choose RE?

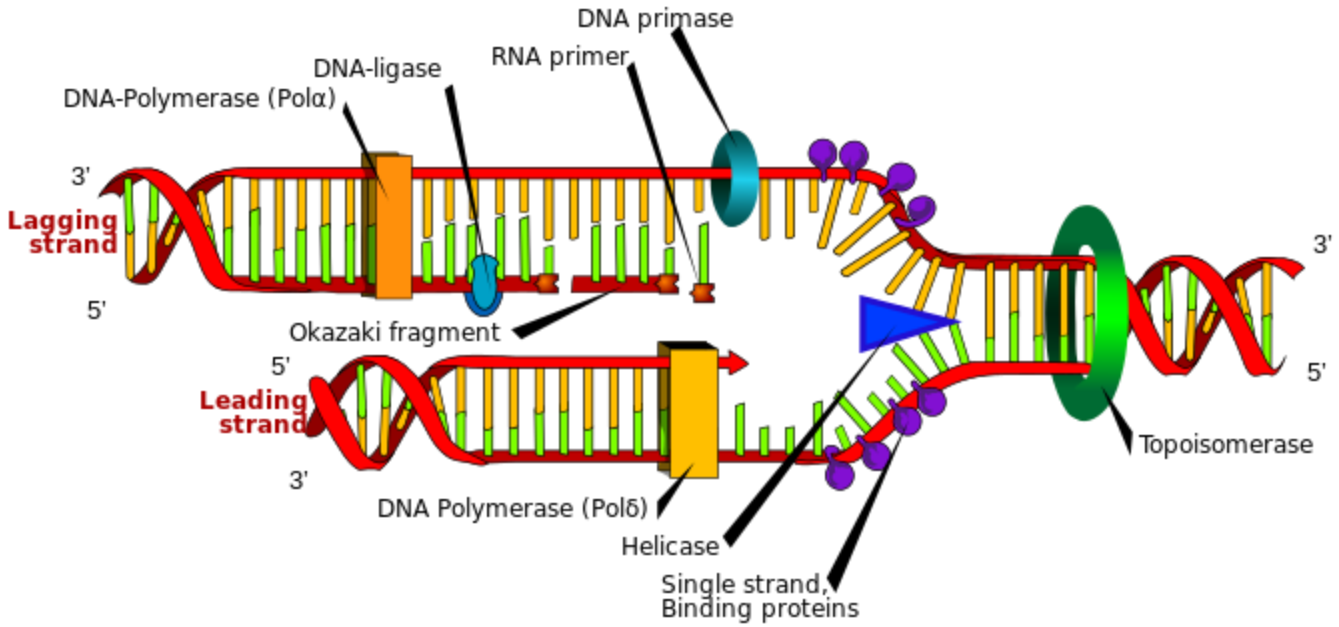
- Type II
- Recognition site in vector?
- Recognition site in insert?
- Easy inactivation

III. Enzymes in cloning

III.2. Ligation enzyme (DNA ligase)

- **In living cells**

  DNA replication and DNA repair



III.2. Ligation enzyme (DNA ligase)

- ***In vitro***

- First extracted in 1967 (from *E. coli*)

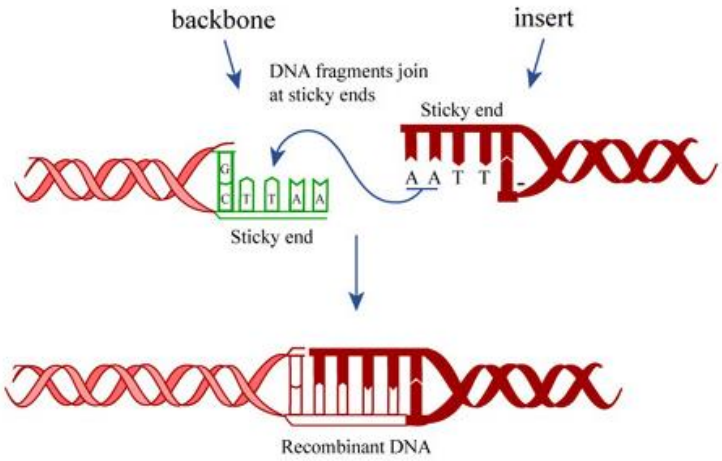
- Types of DNA Ligases

  + *E. coli* DNA ligase

    sticky ends

  + T4 DNA ligase

    both sticky ends and blunt ends

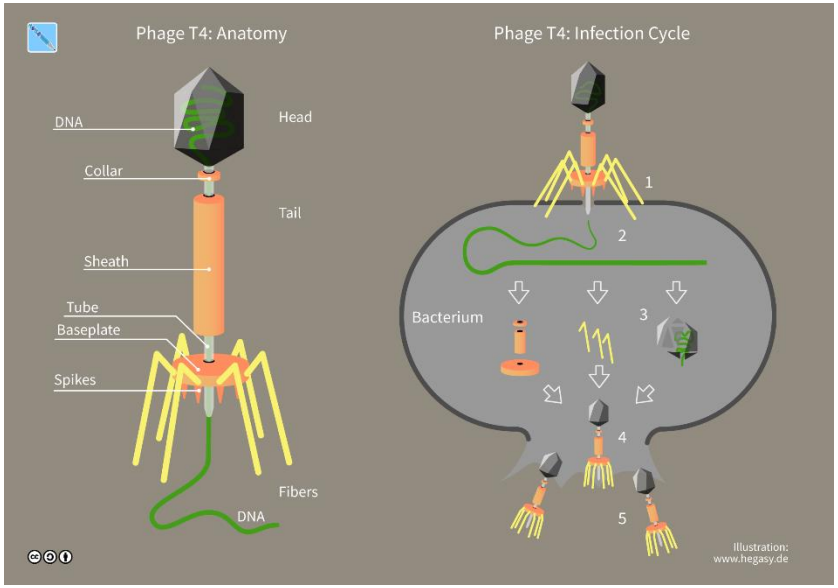


Sticky ends ligate quickly than blunt ends

- 1 U: the amount of DNA ligase required to join 50% of 100 ng of DNA fragments with cohesive termini in 50 µL 1X DNA Ligase Buffer, 30 minute, 23°C (Sigma-Aldrich)

T4 DNA ligase (commercialized)

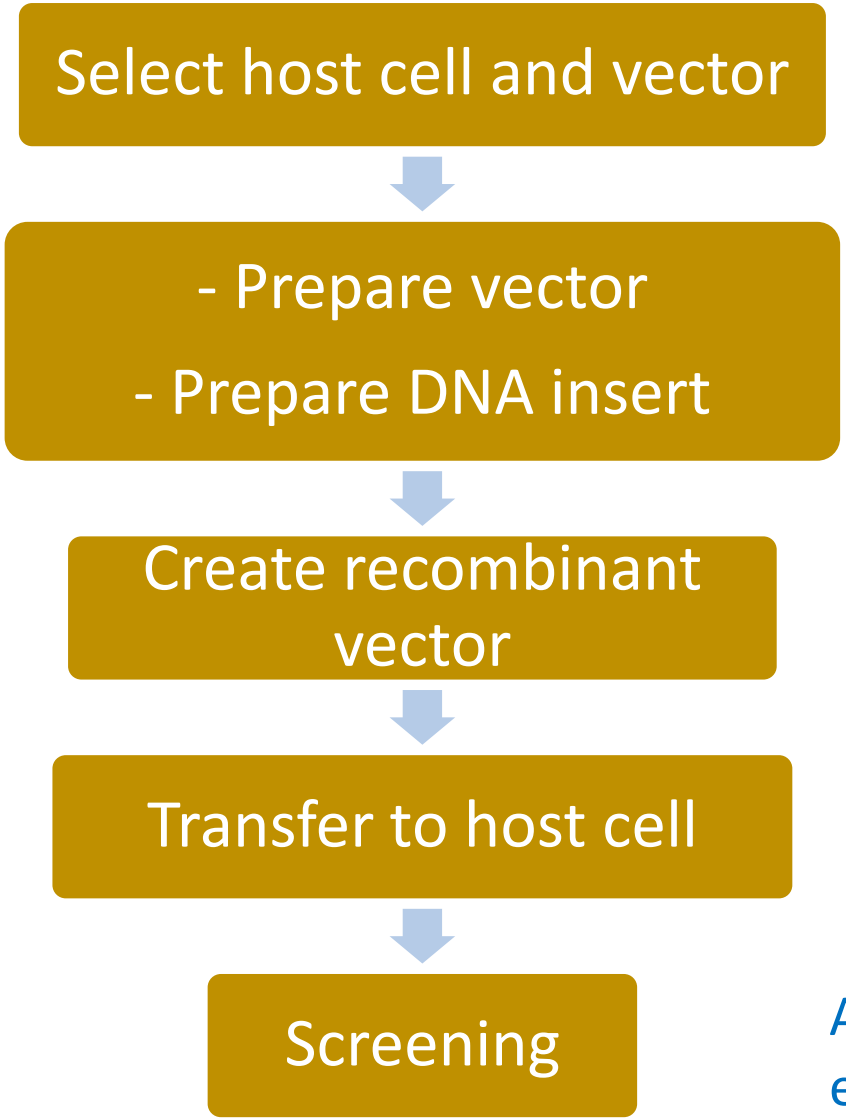
- Source: T4 DNA ligase is encoded by the gene from bacteriophage T4.



- Production: the ligase gene is cloned into a plasmid vector → transformed into a *E. coli* host for overexpression → purification

IV. Cloning process

Cloning steps



Bacteria, yeast, plant cell,…

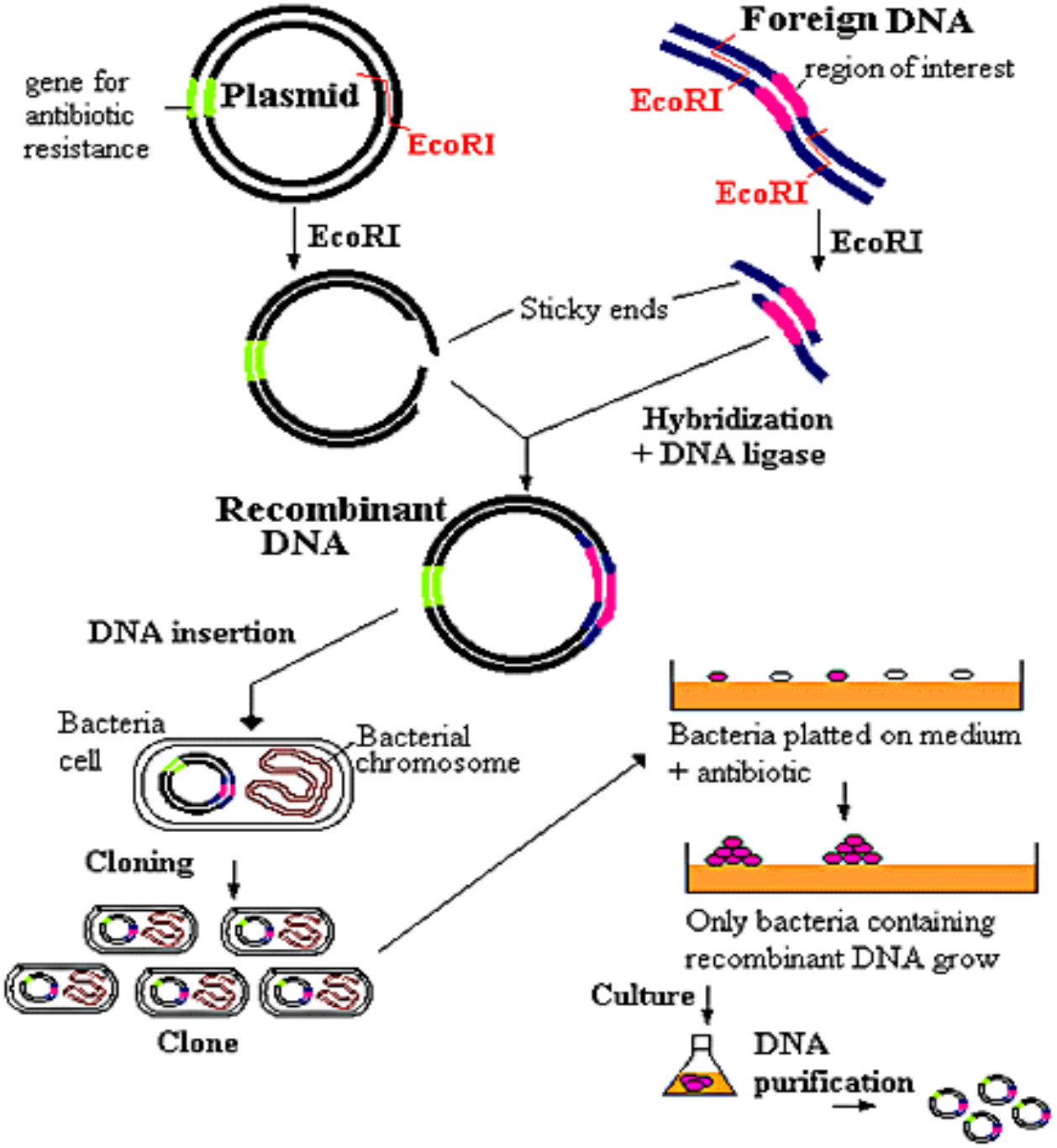
DNA extraction, PCR, purification,….

Cut, ligate, purify,….

Heat shock, electroporation

Antibiotics, colorimetric enzymes,….

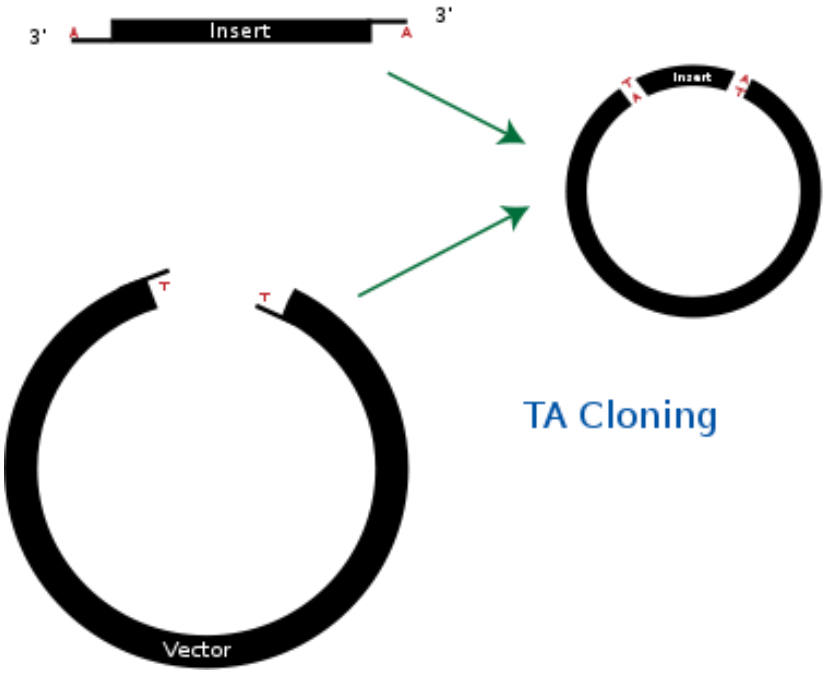
Example


gene for antibiotic resistance
Plasmid
EcoRI
EcoRI
Foreign DNA
region of interest
EcoRI
EcoRI
EcoRI
Sticky ends
Hybridization + DNA ligase
Recombinant DNA
DNA insertion
Bacteria cell
Bacterial chromosome
Cloning
Clone
Bacteria platted on medium + antibiotic
Only bacteria containing recombinant DNA grow
Culture
DNA purification

TA cloning

Principle

- Taq DNA polymerase adds a single adenine (A) overhang to the 3′ ends of PCR products.
- Special vectors are prepared with a single thymine (T) overhang at their ends.
- The complementary A–T base pairing allows the PCR product to ligate into the vector efficiently.



TA cloning

- **Advantages**
  - No need for RE
  - Direct ligation
- **Disadvantages**
  - Dependent on Taq polymerase
  - Non-directional insertion
  - Vectors expensive